
Symmetries in Physics — Exercise Sheet 3

Representations and complete reducibility

Variant: Questions only

Exercise 1. Let $C_3 = \langle r \rangle$ act on itself by left multiplication, and number the basis of its regular representation as $e_1 = e$, $e_2 = e_r$, $e_3 = e_{r^2}$.

- (i) Write the matrices of the operators $\rho(e)$, $\rho(r)$, $\rho(r^2)$ in this basis.
- (ii) Decompose the representation explicitly into one-dimensional representations, exhibiting the change of basis, and say which of the χ_k of Lecture 3 occur.

Exercise 2.

- (i) Find all one-dimensional representations of D_n for $n \geq 3$. How many are there when n is odd, and how many when n is even?
- (ii) Give the kernel of each.
- (iii) Show that none of them is faithful.

Exercise 3. Let D_4 act on the plane by its defining representation, with r the rotation through $\pi/2$ and $s(x_1, x_2) = (x_1, -x_2)$ the reflection in the x_1 -axis. Consider the function-space representation

$$(\rho(g)f)(\vec{r}) = f(g^{-1}\vec{r})$$

on the three-dimensional space of quadratic polynomials spanned by x_1^2 , x_1x_2 , x_2^2 .

- (i) Write the matrices of the operators $\rho(r)$ and $\rho(s)$ in this basis.
- (ii) Decompose the space into three invariant lines, and say which one-dimensional representation of Exercise 2 each line carries.
- (iii) Which one-dimensional representation of D_4 does not occur?

Exercise 4. A remark at the end of Lecture 4 claims that the standard representation of S_3 and the defining representation of D_3 are equivalent. Prove it. Use the matrices $B(g)$ of the worked example, in the basis $f_1 = e_1 - e_2$, $f_2 = e_2 - e_3$, the matrices $D(r)$, $D(s)$ of the defining representation, and the isomorphism $r \mapsto (1\ 2\ 3)$, $s \mapsto (2\ 3)$ of Lecture 1.

- (i) Compute $B((2\ 3))$.
- (ii) Find an invertible T with $T B((1\ 2\ 3)) = D(r) T$ and $T B((2\ 3)) = D(s) T$, and explain why checking the two generators is enough.

Hint: send e_k to vertex k of the triangle.

Exercise 5. Let $C_2 = \{e, p\}$ act on wave functions $\psi \in L^2(\mathbb{R})$ by the parity operator $(\Pi\psi)(x) = \psi(-x)$, where $\Pi = \rho(p)$.

- (i) Show that this is the function-space representation of the action $p \cdot x = -x$ of C_2 on $\mathbb{R}$.
- (ii) Show directly that the $+1$ and -1 eigenspaces of Π are invariant and are precisely the even and odd functions.
- (iii) Prove that every ψ decomposes uniquely into one function of each kind, and state how p acts on the two invariant pieces.

Exercise 6. Let $C_3 = \langle r \rangle$ act on $\mathbb{C}^2$ by $A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$.

- (i) Check $A^3 = 1$, and show that A is not unitary for the standard inner product.
- (ii) Average the standard inner product over the group, and check that the result M satisfies $A^\dagger M A = M$.
- (iii) A is the matrix $B((1\ 2\ 3))$ of Lecture 4's worked example. Compare M with the Gram matrix of $f_1 = e_1 - e_2$ and $f_2 = e_2 - e_3$ in $\mathbb{C}^3$, and explain the result without computing.
- (iv) Find a basis, orthonormal for M , in which A is the rotation matrix $R(2\pi/3)$.

Exercise 7. Maschke's theorem has two hypotheses. This exercise shows what each is for.

- (i) Take the representation $D(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ of the infinite group $(\mathbb{Z}, +)$ from Lecture 4's example "Reducible but indecomposable". Which step of the averaging proof cannot be carried out? Prove that no inner product on $\mathbb{C}^2$ makes ρ unitary.
- (ii) Now let the numbers be *bits*: vectors in $V = \{0, 1\}^2$, added and scaled modulo 2. Let $C_2 = \{e, a\}$ act on V by swapping coordinates. Show that the line spanned by $(1, 1)$ is invariant but has no invariant complement, and say where the averaging proof breaks here.